PART A — QUESTIONS

AF21-001
Question:
Solve: $23x - 16 = 122$
A) 5
B) 6
C) 7
D) 8

AF21-002
Question:
If $21n + 6 = 153$, what is n?
A) 5
B) 6
C) 7
D) 8

AF21-003
Question:
Simplify: $8(3x - 4) - 7x$
A) $17x - 32$
B) $24x - 32$
C) $17x - 4$
D) $31x - 32$

AF21-004
Question:
A company charges $45 per inspection plus a $20 scheduling fee. Which expression represents the total charge for i inspections?
A) $45i + 20$
B) $20i + 45$
C) $65i$
D) $45 - 20i$

AF21-005
Question:
If $f(x) = 18x - 52$, what is $f(8)$?
A) 82
B) 92
C) 144
D) 196

AF21-006
Question:
Which ordered pair satisfies the equation $y = 16x - 9$?
A) (1, 8)
B) (2, 22)
C) (3, 39)
D) (4, 56)

AF21-007
Question:
Solve: 25(x - 7) = 150
A) 11
B) 12
C) 13
D) 14

AF21-008
Question:
A pattern begins: 11, 31, 51, 71, ... What is the next number?
A) 89
B) 90
C) 91
D) 92

AF21-009
Question:
Solve: x/45 = 3
A) 48
B) 90
C) 135
D) 180

AF21-010
Question:
A line has a slope of -18 and crosses the y-axis at 15. Which equation represents the line?
A) y = 15x - 18
B) y = -18x + 15
C) y = 18x + 15
D) y = -15x + 18

AF21-011
Question:
If 7 cases contain 266 parts, how many parts are in 4 cases at the same rate?
A) 144
B) 152
C) 160
D) 168

AF21-012
Question:
Solve: 330 - 32x = 202
A) 3
B) 4
C) 5
D) 6

AF21-013
Question:
Which expression is equivalent to 105m - 48m - 20?

A) 57m - 20
B) 153m - 20
C) 57m + 20
D) 105m - 68

AF21-014
Question:
A supply shelf starts with 1,750 parts. Each kit uses 125 parts. Which equation gives the number of parts P left after k kits?
A) P = 1,750k - 125
B) P = 125k + 1,750
C) P = 1,750 - 125k
D) P = 125k - 1,750

AF21-015
Question:
If y = -24x + 350, what is y when x = 13?
A) 28
B) 38
C) 312
D) 662

AF21-016
Question:
Solve: 36x - 72 = 24x + 72
A) 10
B) 11
C) 12
D) 13

AF21-017
Question:
Which value of x makes 20x - 8 greater than 232?
A) 11
B) 12
C) 13
D) 10

AF21-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: -10, 11, 32, 53

What is the rule?
A) y = 21x - 10
B) y = 10x + 21
C) y = -10x + 21
D) y = 21x + 10

AF21-019
Question:
Solve: $20(x + 3) = 17x + 96$
A) 10
B) 11
C) 12
D) 13

AF21-020
Question:
If $a = -16$ and $b = 13$, what is $4a + 5b$?
A) -129
B) -1
C) 1
D) 129

AF21-021
Question:
Which point is located in Quadrant I?
A) (-18, 7)
B) (18, -7)
C) (-18, -7)
D) (18, 7)

AF21-022
Question:
A worker labels 264 boxes in 12 minutes. At the same rate, how many boxes can the worker label in 5 minutes?
A) 100
B) 110
C) 120
D) 132

AF21-023
Question:
Solve for w: $E = 36w + 72$
A) $w = E - 72$
B) $w = (E - 72)/36$
C) $w = 36(E - 72)$
D) $w = E/36 + 72$

AF21-024
Question:
What is the slope of a line that goes through (9, 18) and (15, 102)?
A) 12
B) 13
C) 14
D) 84

AF21-025
Question:

Simplify: $84x + 25 - 73x - 43$
A) $11x - 18$
B) $157x - 18$
C) $11x + 68$
D) $157x + 68$

AF21-026
Question:
If $g(x) = x^2 - 12x$, what is $g(16)$?
A) 48
B) 64
C) 192
D) 256

AF21-027
Question:
A sequence follows the rule "add 14, then multiply by 4." If the first term is 7, what is the second term?
A) 21
B) 49
C) 84
D) 98

AF21-028
Question:
Solve: $25(x - 4) + 30 = 255$
A) 11
B) 12
C) 13
D) 14

AF21-029
Question:
A line crosses the y-axis at -23 and rises 11 units for every 1 unit to the right. Which equation represents the line?
A) $y = -23x + 11$
B) $y = 11x - 23$
C) $y = -11x - 23$
D) $y = 23x + 11$

AF21-030
Question:
If $30/45 = x/135$, what is x?
A) 75
B) 80
C) 90
D) 105

AF21-031
Question:
Solve: $-40x + 200 = -280$

A) -12
B) -8
C) 8
D) 12

AF21-032
Question:
A warehouse starts with 6,000 fasteners. Each job uses 300 fasteners. Which expression gives the number of fasteners left after j jobs?
A) 6,000 + 300j
B) 6,000 - 300j
C) 300j - 6,000
D) 6,000 ÷ 300j

AF21-033
Question:
Which equation has x = 28 as its solution?
A) 2x + 18 = 72
B) 5x - 35 = 105
C) x/14 + 9 = 12
D) 3x - 17 = 70

PART B — ANSWER KEY + EXPLANATIONS

AF21-001 — Correct: B — Explanation:
Solve 23x - 16 = 122. Add 16 to both sides: 23x = 138. Divide by 23: x = 6. The most common mistake is subtracting 16 instead of adding 16.

AF21-002 — Correct: C — Explanation:
Start with 21n + 6 = 153. Subtract 6 from both sides: 21n = 147. Divide by 21: n = 7. A common mistake is stopping at 147 and forgetting to divide by 21.

AF21-003 — Correct: A — Explanation:
Distribute first: 8(3x - 4) = 24x - 32. Then subtract 7x: 24x - 7x - 32 = 17x - 32. A common mistake is forgetting to multiply -4 by 8.

AF21-004 — Correct: A — Explanation:
The company charges $45 for each inspection, so that part is 45i. The $20 scheduling fee is added one time. The total charge is 45i + 20. A common mistake is treating the scheduling fee as the per-inspection charge.

AF21-005 — Correct: B — Explanation:
Substitute 8 for x: f(8) = 18(8) - 52 = 144 - 52 = 92. A common mistake is adding 52 instead of subtracting it.

AF21-006 — Correct: C — Explanation:
Test the choices in y = 16x - 9. For (3, 39), 16(3) - 9 = 48 - 9 = 39, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF21-007 — Correct: C — Explanation:
Solve $25(x - 7) = 150$. Divide both sides by 25: $x - 7 = 6$. Add 7: $x = 13$. A common mistake is adding 7 before undoing the multiplication by 25.

AF21-008 — Correct: C — Explanation:
The pattern increases by 20 each time: 11, 31, 51, 71. Add 20 to 71: 91. A common mistake is assuming the increase is 19 because the numbers are close together.

AF21-009 — Correct: C — Explanation:
$x/45 = 3$ means x divided by 45 equals 3. Multiply both sides by 45: $x = 135$. A common mistake is adding 45 and 3 instead of multiplying.

AF21-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -18 and the y-intercept is 15, so the equation is $y = -18x + 15$. A common mistake is switching the slope and y-intercept.

AF21-011 — Correct: B — Explanation:
Find the unit rate: 266 parts ÷ 7 cases = 38 parts per case. For 4 cases: $38 \times 4 = 152$. A common mistake is multiplying 266 by 4 without finding the rate first.

AF21-012 — Correct: B — Explanation:
Solve $330 - 32x = 202$. Subtract 330 from both sides: $-32x = -128$. Divide by -32: $x = 4$. A common mistake is losing the negative sign after subtracting 330.

AF21-013 — Correct: A — Explanation:
Combine like terms: $105m - 48m = 57m$. The constant -20 stays the same. The expression becomes $57m - 20$. A common mistake is combining the constant with the variable terms.

AF21-014 — Correct: C — Explanation:
The shelf starts with 1,750 parts. Each kit uses 125 parts, so k kits use 125k parts. The number left is $P = 1{,}750 - 125k$. A common mistake is adding 125k even though parts are being used.

AF21-015 — Correct: B — Explanation:
Substitute $x = 13$ into $y = -24x + 350$: $y = -24(13) + 350 = -312 + 350 = 38$. A common mistake is treating $-24(13)$ as positive 312.

AF21-016 — Correct: C — Explanation:
Solve $36x - 72 = 24x + 72$. Subtract 24x: $12x - 72 = 72$. Add 72: $12x = 144$. Divide by 12: $x = 12$. A common mistake is adding 72 to only one side.

AF21-017 — Correct: C — Explanation:
Solve $20x - 8 > 232$. Add 8: $20x > 240$. Divide by 20: $x > 12$. The only listed value greater than 12 is 13. A common mistake is choosing 12, but $20(12) - 8 = 232$, not greater than 232.

AF21-018 — Correct: A — Explanation:
The y-values increase by 21 each time x increases by 1, so the slope is 21. When $x = 0$, $y = -10$, so the y-intercept is -10. The rule is $y = 21x - 10$. A common mistake is using +10 instead of -10.

AF21-019 — Correct: C — Explanation:
Distribute: $20x + 60 = 17x + 96$. Subtract 17x: $3x + 60 = 96$. Subtract 60: $3x = 36$. Divide by 3: $x = 12$. A common mistake is forgetting to multiply 3 by 20.

AF21-020 — Correct: C — Explanation:
Substitute a = -16 and b = 13: 4a + 5b = 4(-16) + 5(13) = -64 + 65 = 1. A common mistake is treating 4(-16) as positive 64.

AF21-021 — Correct: D — Explanation:
Quadrant I has a positive x-coordinate and a positive y-coordinate. The point (18, 7) fits this description. A common mistake is choosing a point with only one positive coordinate.

AF21-022 — Correct: B — Explanation:
264 boxes in 12 minutes means 264 ÷ 12 = 22 boxes per minute. In 5 minutes: 22 × 5 = 110. A common mistake is multiplying 264 by 5 without finding the rate first.

AF21-023 — Correct: B — Explanation:
Start with E = 36w + 72. Subtract 72 from both sides: E - 72 = 36w. Divide by 36: w = (E - 72)/36. A common mistake is dividing E by 36 first and then subtracting 72.

AF21-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (9, 18) to (15, 102), y increases by 84 and x increases by 6. The slope is 84 ÷ 6 = 14. A common mistake is using 84 as the slope without dividing by the change in x.

AF21-025 — Correct: A — Explanation:
Combine like terms: 84x - 73x = 11x, and 25 - 43 = -18. The simplified expression is 11x - 18. A common mistake is writing 11x + 18 because the negative sign before 43 is missed.

AF21-026 — Correct: B — Explanation:
Substitute 16 for x: $g(16) = 16^2 - 12(16) = 256 - 192 = 64$. A common mistake is calculating 16^2 as 32 instead of 256.

AF21-027 — Correct: C — Explanation:
Start with 7. Add 14: 21. Then multiply by 4: 84. A common mistake is multiplying first and then adding 14.

AF21-028 — Correct: C — Explanation:
Solve 25(x - 4) + 30 = 255. Distribute: 25x - 100 + 30 = 255. Combine: 25x - 70 = 255. Add 70: 25x = 325. Divide by 25: x = 13. A common mistake is forgetting to combine -100 and +30.

AF21-029 — Correct: B — Explanation:
The line rises 11 units for every 1 unit to the right, so the slope is 11. It crosses the y-axis at -23, so the equation is y = 11x - 23. A common mistake is switching the slope and intercept.

AF21-030 — Correct: C — Explanation:
Use the proportion 30/45 = x/135. Since 45 × 3 = 135, multiply 30 by 3: x = 90. A common mistake is adding 90 to the numerator because 45 becomes 135.

AF21-031 — Correct: D — Explanation:
Solve -40x + 200 = -280. Subtract 200 from both sides: -40x = -480. Divide by -40: x = 12. A common mistake is giving -12 because both sides contain negative numbers.

AF21-032 — Correct: B — Explanation:
The warehouse starts with 6,000 fasteners. Each job uses 300 fasteners, so j jobs use 300j fasteners.

The number left is 6,000 - 300j. A common mistake is adding 300j even though fasteners are being used.

AF21-033 — Correct: B — Explanation:
Test x = 28. In choice B, 5(28) - 35 = 140 - 35 = 105, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.